

Homework 6: Ordination

S&DS 5360 | Multivariate Statistics

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Setup & Data Overview

We continue our work with the Healthy Minds Survey (HMS) data. For Correspondence ANalysis and NMDS, we will select related survey questions measured on the same scale. For these, we select:

- Depression (`phq9_1` to `phq9_9`): Nine questions on interest, mood, sleep, and energy levels over 2 weeks.
- Anxiety (`gad7_1` to `gad7_7`): Seven questions on nervousness, worrying, and restlessness over 2 weeks.

For Canonical Correspondence Analysis (CCA) and non-linear wireplots, we select continuous variables that may explain the variation in symptoms:

- Age (`age`): Continuous measure of respondent's age.
- Loneliness Score (`lonesc`): Sum of UCLA Loneliness Scale (3–9)
- Exercise Hours (`exerc`): Average hours per week spent exercising (1–5 scale).
- Internet Use (`internet_1`): Time spent online (1–8).

We omit the code to preprocess the data.

Parts 1–4 | Correspondence Analysis

We begin by performing an indirect ordination to explore the underlying structure of mental health symptoms among students, particularly to reduce the dimensionality of our 16 symptom variables across PHQ-9 and GAD-7 to more easily visualize how they group together. To do so, we complete a Correspondence Analysis, which is ideal for our data because the symptom data represents frequency counts (scales from 1 = “Not at All” to 4 = “Nearly Every Day”). CA maps both students and symptoms into low dimensional space via chi-square distances.

```
hms_ca <- cca(hms_symptoms)
summary(hms_ca)
```

Call:

```
cca(X = hms_symptoms)
```

Partitioning of scaled Chi-square:

	Inertia	Proportion
Total	0.1073	1
Unconstrained	0.1073	1

Eigenvalues, and their contribution to the scaled Chi-square

Importance of components:

	CA1	CA2	CA3	CA4	CA5	CA6
Eigenvalue	0.01769	0.01133	0.01063	0.008399	0.007819	0.007648
Proportion Explained	0.16484	0.10561	0.09905	0.078261	0.072859	0.071266
Cumulative Proportion	0.16484	0.27045	0.36950	0.447762	0.520621	0.591888

	CA7	CA8	CA9	CA10	CA11	CA12
Eigenvalue	0.006843	0.006118	0.005982	0.005109	0.004899	0.004552
Proportion Explained	0.063769	0.057013	0.055737	0.047603	0.045653	0.042412
Cumulative Proportion	0.655657	0.712670	0.768407	0.816010	0.861663	0.904075

	CA13	CA14	CA15
Eigenvalue	0.003909	0.003677	0.002708
Proportion Explained	0.036422	0.034268	0.025235
Cumulative Proportion	0.940497	0.974765	1.000000

We see that our total inertia is 0.1073, representing the total variance in our data (calculated as a scaled chi-square statistic). Axis 1 (CA1) explains 16.48% of the variance, while Axis 2 (CA2) explains 10.56%, or together, 27.04% of the variance in total.

Observe on our two-dimensional plot of Axes 1 and 2 that our clustered symptoms in blue text tightly overlap near the origin (0,0). Because there are so many students pulling the axes outward, the variation among symptoms themselves make them appear clumped in the middle. We provide a zoomed-in view on the right-hand side to more easily discern the symptoms.

```
par(mfrow=c(1,2))

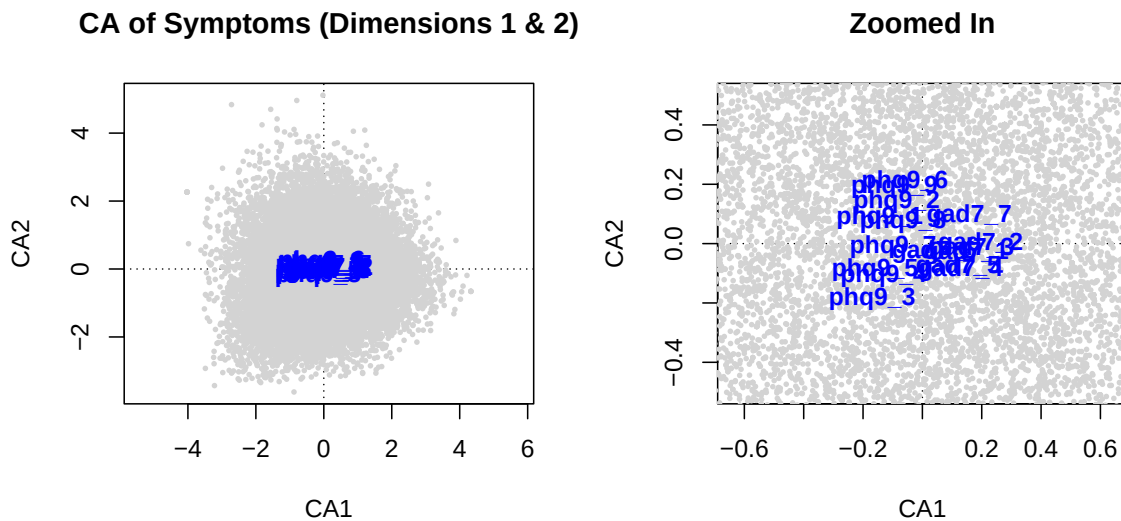
# Full Plot
plot(hms_ca, choices = c(1, 2), type = "n",
     main = "CA of Symptoms (Dimensions 1 & 2)")
```

```

points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)

# Zoomed-In Plot
plot(hms_ca, choices = c(1, 2), type = "n",
      xlim = c(-.5, .5), ylim = c(-.5, .5),
      main = "Zoomed In")
points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)

```



Upon visualizing the first two dimensions of the Correspondence Analysis, we see there is no clear evidence of an arch effect (data snaking). Rather than forming a classic U or V crescent shape, the data forms a dense, continuous cloud centered around the origin, with the symptom variables tightly clustered in the middle.

As discussed in class, this lack of snaking would be expected when there is complete species turnover along an environmental gradient. Our mental health symptoms do not behave this way; rather than “turning over,” symptoms are highly correlated and additive. For example, a student experiencing severe depression does not replace their more mild symptoms (like fatigue) with more severe ones (like self harm). Rather, they may experience both simultaneously. Since these symptoms co-occur, the CA maps them as the tightly inter-correlated cluster we see in the plots.

Given the above, we conclude that mental health symptoms measured by PHQ-9 and GAD-7 are highly comorbid and do not represent distinct, mutually exclusive conditions among the student population. The lack of data snaking and the tight clustering of symptom variables

near the origin indicate that students do not experience a turnover symptoms. Instead, the first two axes, which capture $\approx 27\%$ of the variance, likely represent a single, dominant gradient of overall psychological distress. Students scale along this gradient additively, experiencing these anxiety and depression symptoms simultaneously rather than in isolation of each other.

Parts 5–7 | Multidimensional Scaling

Now, we perform Non-Metric Multidimensional Scaling (NMDS) in one, two, and three dimensions. Unlike Correspondence Analysis, which relies on chi-square distances, NMDS's rank-based approach maps respondents into a low-dimensional space such that the distances between points optimally preserve the rank-order of their actual dissimilarities in the full 16-dimensional symptom space.

We elect to use the Bray-Curtis distance metric (as opposed to Euclidean distance) because of its robustness to joint absences. For example, if two students both select “not at all” for a number of severe symptoms, Bray-Curtis does not treat this shared absence as a strong indicator of similarity, focusing instead on the actually experienced symptoms. This is optimal for survey data like ours.

```
set.seed(4747)

df_subset <- df %>% sample_n(500)

hms_symptoms_subset <- df_subset %>% select(all_of(symptoms))
hms_env_subset <- df_subset %>% select(all_of(environment))

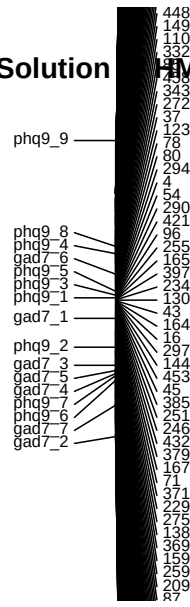
hms_nmds1 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 1, trymax = 50)
hms_nmds2 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 2, trymax = 50)
hms_nmds3 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 3, trymax = 50)
```

Note that we manually set `trymax = 50` to ensure the algorithm incorporates enough random starts to reliably find the global optimum within a reasonable number of runs. Furthermore, we sample 500 rows from our data, as our dataset containing 33,100 rows of symptom data is computationally expensive and unnecessary for ordination.

```
par(mfrow = c(1,1))

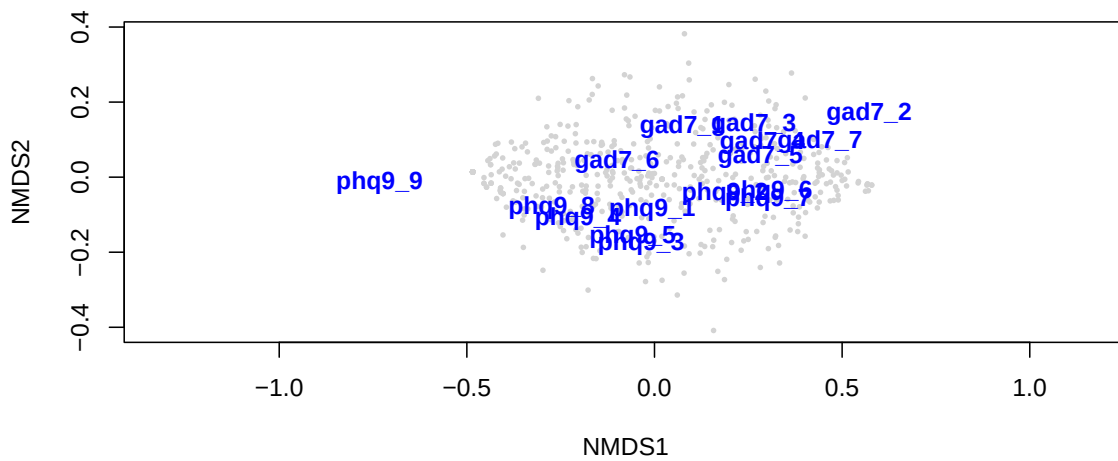
# 1D
plot(hms_nmds1, main = "1D NMDS Solution of HMS Symptoms")
```

1D NMDS Solution of HMS Symptoms



```
# 2D
plot(hms_nmds2, type = "n", main = "2D NMDS Solution of HMS Symptoms")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 1, font = 2)
```

2D NMDS Solution of HMS Symptoms

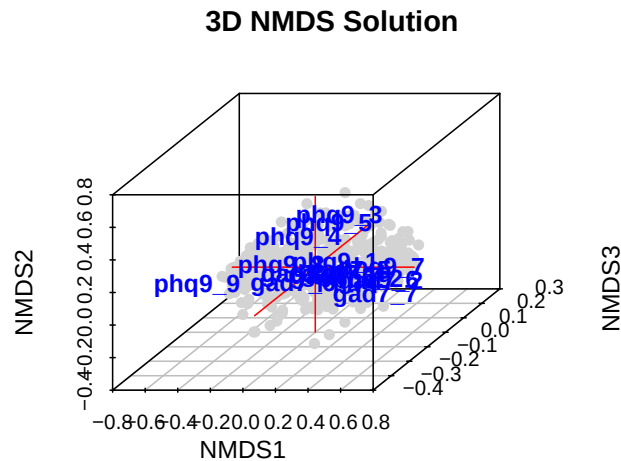


```
# 3D
pl <- ordiplot3d(hms_nmds3, display = "sites", pch = 16, col = "lightgrey",
```

```

main = "3D NMDS Solution")
sp <- scores(hms_nm3, choices = 1:3, display = "species")
text(pl$xyz.convert(sp), rownames(sp), col = "blue", font = 2, cex = 1)

```



In the above plots, we see: - **1D NMDS**: The 1-dimensional plot forces all symptom variance onto a single axis, resulting in an overcrowded plot from which it is difficult to discern meaningful relationships between individual symptoms. - **2D NMDS**: The 2-dimensional plot reveals a much clearer mapping of the symptom space. Observe the extreme observation **phq9_9** on the far left-hand side; this question specifically measures suicidal ideation and self-harm. The remaining general anxiety and depression symptoms form a looser cluster on the right. The visual separation makes sense: self harm is distinctly different from other, less severe depressive symptoms and requires acute care. - **3D NMDS**: The 3-dimensional plot adds an additional depth axis which is not immediately helpful. While it helps pull apart the main cluster of symptoms that overlap in 2D, the overall topology remains the same, with **phq9_9** retaining the same distinct separation.

```

cat("Stress for 1D:", hms_nm3$stress, "\n")

```

Stress for 1D: 0.2000061

```

cat("Stress for 2D:", hms_nm3$stress, "\n")

```

Stress for 2D: 0.1346835

```
cat("Stress for 3D:", hms_nmds3$stress, "\n")
```

Stress for 3D: 0.1057252